

Address of Code, Global, Static, Heap and Stack

Question

Write a C program to display the addresses of code, global, static, heap and stack variables and identify their respective regions in the Linux process address space.

```

    printf("\nProcess ID: %d\n", getpid());

    free(heap_variable);

    return 0;
}
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat Practical7B.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_variable = 100;

int main() {

    int stack_variable = 200;

    int *heap_variable =
        malloc(sizeof(int));

    if (heap_variable == NULL) {
        return 1;
    }

    *heap_variable = 300;

    printf("Process ID: %d\n", getpid());

    printf("Stack address: %p\n",
        (void *)&stack_variable);

    printf("Heap address : %p\n",
        (void *)heap_variable);

```

```
#include <unistd.h>

int global_variable = 100;

static int static_variable = 200;

void sample_function() {
    printf("Inside sample function.\n");
}

int main() {

    int stack_variable = 300;

    int *heap_variable;

    heap_variable = malloc(sizeof(int));

    if (heap_variable == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    *heap_variable = 400;

    printf("\nPROCESS ADDRESS SPACE\n\n");

    printf("Code address    : %p\n",
           (void *)sample_function);

    printf("Global address : %p\n",
           (void *)&global_variable);

    printf("Static address : %p\n",
           (void *)&static_variable);
```

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano Practical7A.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc Practical7A.c -o Practical7A
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./Practical7A
```

PROCESS ADDRESS SPACE

```
Code address   : 0x636c899cc1e9
Global address  : 0x636c899cf010
Static address  : 0x636c899cf014
Heap address    : 0x636cb029b010
Stack address   : 0x7fff8cf2eb5c
```

Process ID: 1021

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano Practical7B.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc Practical7B.c -o Practical7B
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./Practical7B
```

Process ID: 2664

```
Stack address: 0x7ffdc3c8fcfc
Heap address  : 0x6105bb7cf010
```

Process is running.

Open another terminal and execute:

```
cat /proc/2664/maps
```

Press Enter to exit...

^C

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat /proc/5678/maps
cat: /proc/5678/maps: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat /proc/2664/maps
cat: /proc/2664/maps: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat /proc/2664/maps
cat: /proc/2664/maps: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat /proc/1021/maps
cat: /proc/1021/maps: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat Practical7A.c
```

```
printf("\nProcess is running.\n");  
  
printf("Open another terminal and execute:\n");  
  
printf("cat /proc/%d/maps\n", getpid());  
  
printf("\nPress Enter to exit...\n");  
  
getchar();  
  
free(heap_variable);  
  
return 0;
```

```
ikrishna__90156@DESKTOP-2JF9H5T:~$ |
```